Classification – business perspective

Lift > Gain > Profit

The idea of a predictive model

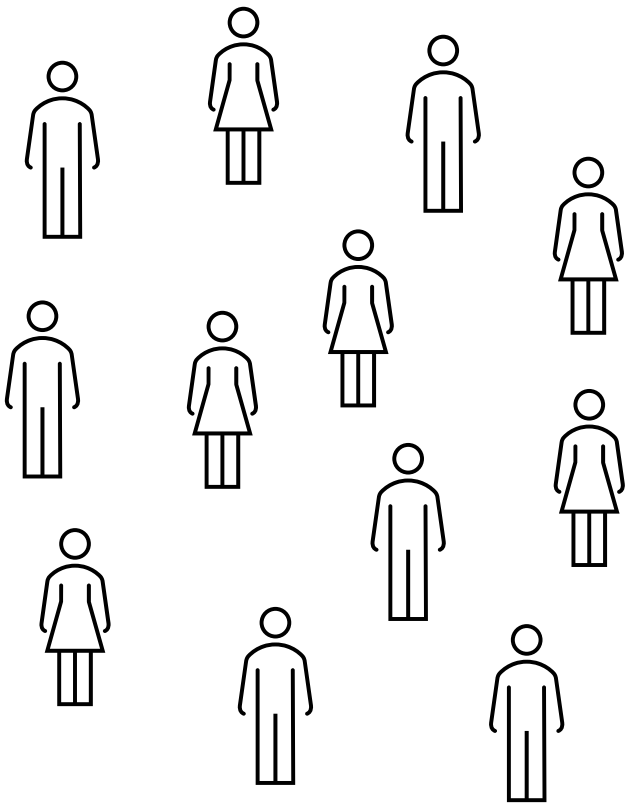
We have some customers

We want to contact them and convince to something

This "convincing" costs money, but also generates profit

The problem: only some of them will respond to our offer.

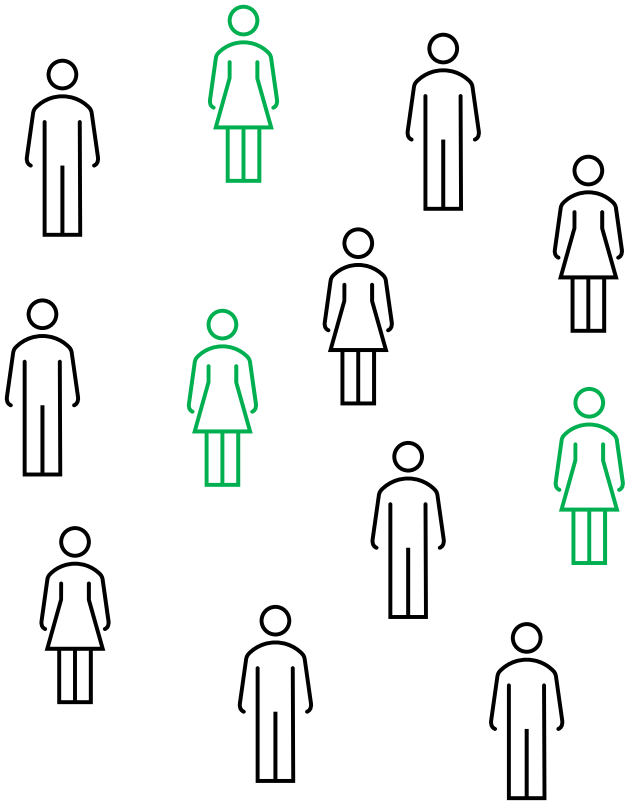
How to find them?



The idea of a predictive model

We can contact them randomly, or with some "sense"

"Sense" means a higher chance of a customer response, such as a purchase.



The idea of a predictive model

Na szczęście, dysponujemy historią naszych doświadczeń z klientami: ich cechy oraz reakcję na naszą ofertę

Customer	Anwered?
Zosia	1
Kasia	0
Kasia	0
Tomek	1
Adam	0
Ireneusz	0
Bartek	0
Andrzej	1
Adam	0
Hania	0
Andrzej	1
Hania	0
Bartek	0
Ireneusz	0
Zosia	0
Kasia	0
Patrycja	1
Patrycja	1
Ireneusz	0
Ewa	0
Hania	0
Bartek	0
Ewa	0
Adam	0
Zosia	1
Ewa	0
Patrycja	1
Andrzej	0
Tomek	1
Tomek	0

The idea of a predictive model

This allows us to:

1. Train a model on this data....

2. ... which will assign new customers the probability of responding to our offer

3. ... which in turn, after sorting according to these probabilities, will indicate those who should be contacted first

Customer	Answered (in real)?	Model prediction (probability)
Tomek	1	0,91
Andrzej	1	0,91
Andrzej	1	0,78
Tomek	0	0,70
Adam	0	0,67
Tomek	1	0,66
Patrycja	1	0,61
Kasia	0	0,56
Zosia	1	0,56
Patrycja	1	0,55
Ireneusz	0	0,54
Bartek	0	0,54
Ireneusz	0	0,54
Bartek	0	0,53
Patrycja	1	0,52
Hania	0	0,52
Adam	0	0,45
Zosia	1	0,38
Ewa	0	0,37
Andrzej	0	0,34
Ireneusz	0	0,27
Hania	0	0,23
Bartek	0	0,23
Ewa	0	0,21
Kasia	0	0,21
Zosia	0	0,12
Hania	0	0,12
Adam	0	0,12
Kasia	0	0,11
Ewa	0	0,08

How to assess the value of such a model?

Can this be done prior to deployment?

Fortunately, yes: after all, we have historical (training) data.

We can ask the model to make a prediction, order the customers by probability, and then see what would happen if we contacted the customers indicated by the model in the first place.

And compare conversion rates across customer segments (deciles):

without ordering vs with ordering

Customer	Answered (in real)?	Model prediction (probability)
Zosia	1	0,56
Kasia	0	0,56
Kasia	0	0,11
Tomek	1	0,91
Adam	0	0,45
Ireneusz	0	0,27
Bartek	0	0,23
Andrzej	1	0,91
Adam	0	0,67
Hania	0	0,12
Andrzej	1	0,78
Hania	0	0,23
Bartek	0	0,53
Ireneusz	0	0,54
Zosia	0	0,12
Kasia	0	0,21
Patrycja	1	0,55
Patrycja	1	0,61
Ireneusz	0	0,54
Ewa	0	0,21
Hania	0	0,52
Bartek	0	0,54
Ewa	0	0,37
Adam	0	0,12
Zosia	1	0,38
Ewa	0	0,08
Patrycja	1	0,52
Andrzej	0	0,34
Tomek	1	0,66
Tomek	0	0,7

Random order

Customer	Answered (in real)?	Model prediction (probability)
Tomek	1	0,91
Andrzej	1	0,91
Andrzej	1	0,78
Tomek	0	0,70
Adam	0	0,67
Tomek	1	0,66
Patrycja	1	0,61
Kasia	0	0,56
Zosia	1	0,56
Patrycja	1	0,55
Ireneusz	0	0,54
Bartek	0	0,54
Ireneusz	0	0,54
Bartek	0	0,53
Patrycja	1	0,52
Hania	0	0,52
Adam	0	0,45
Zosia	1	0,38
Ewa	0	0,37
Andrzej	0	0,34
Ireneusz	0	0,27
Hania	0	0,23
Bartek	0	0,23
Ewa	0	0,21
Kasia	0	0,21
Zosia	0	0,12
Hania	0	0,12
Adam	0	0,12
Kasia	0	0,11
Ewa	0	0,08

Sorted by probability

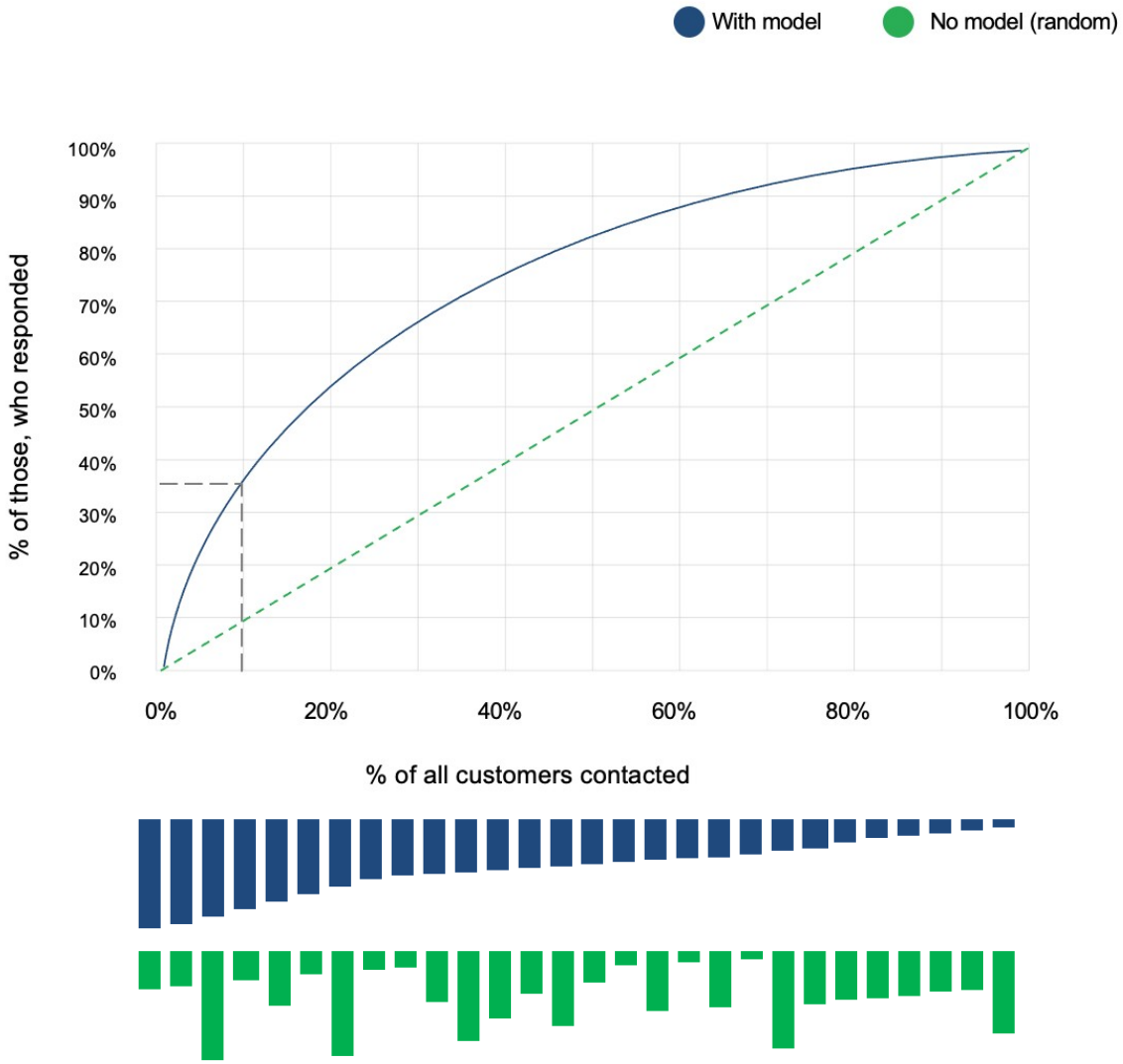
Lift for 10% of cases

In our example:
Case = customer
Positive case = customer who responded to the offer

10% of cases = 1 decile

Without model: 10% of positive cases
With model: 35% of positive cases

Lift = 0.35/0.10 = 3.5

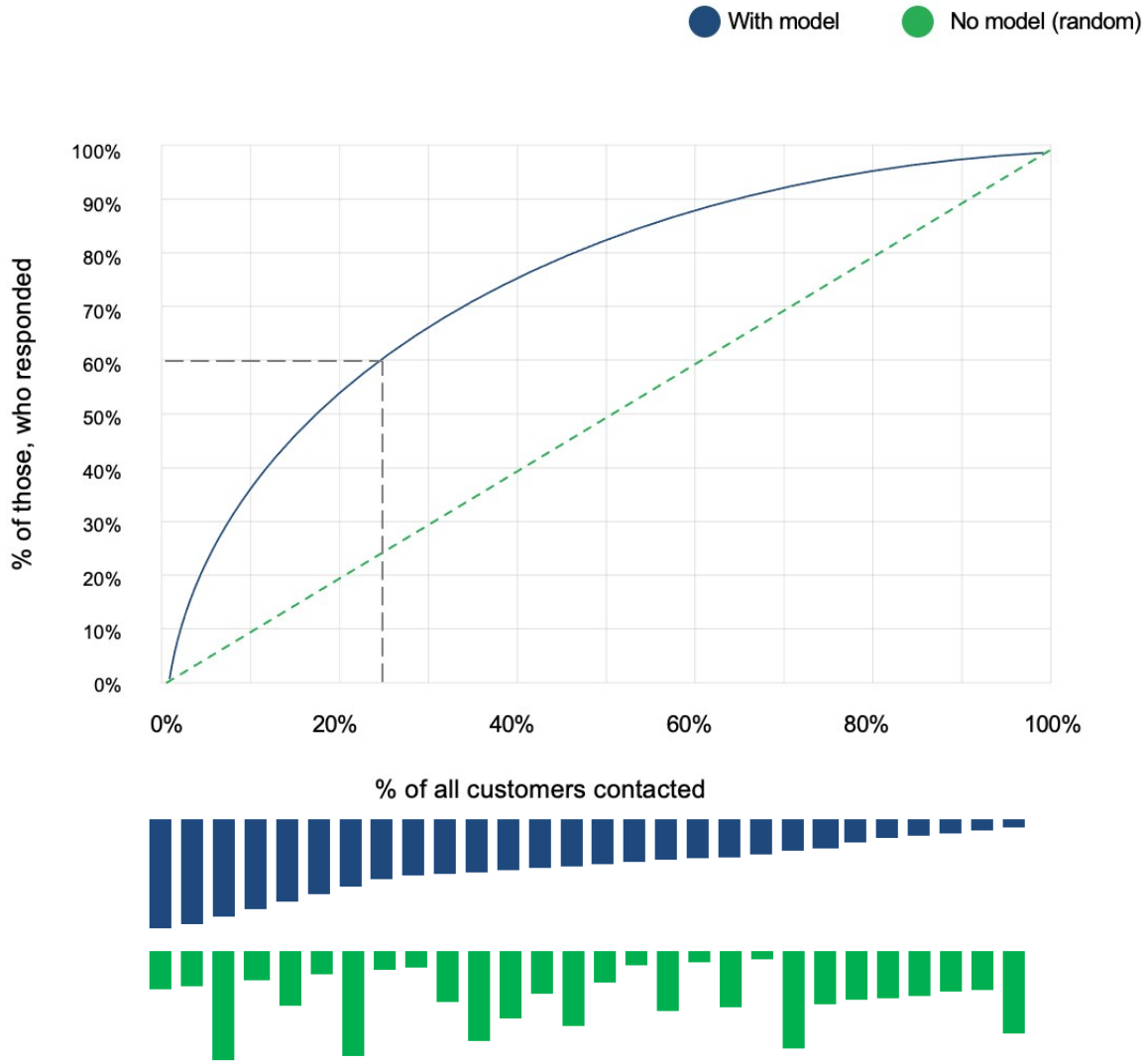


Lift for 25% of cases

25% of cases = 1st quartile

Without model: 25% of positive cases
With model: 60% of positive cases

Lift = 0.6/0.25 = 2.4

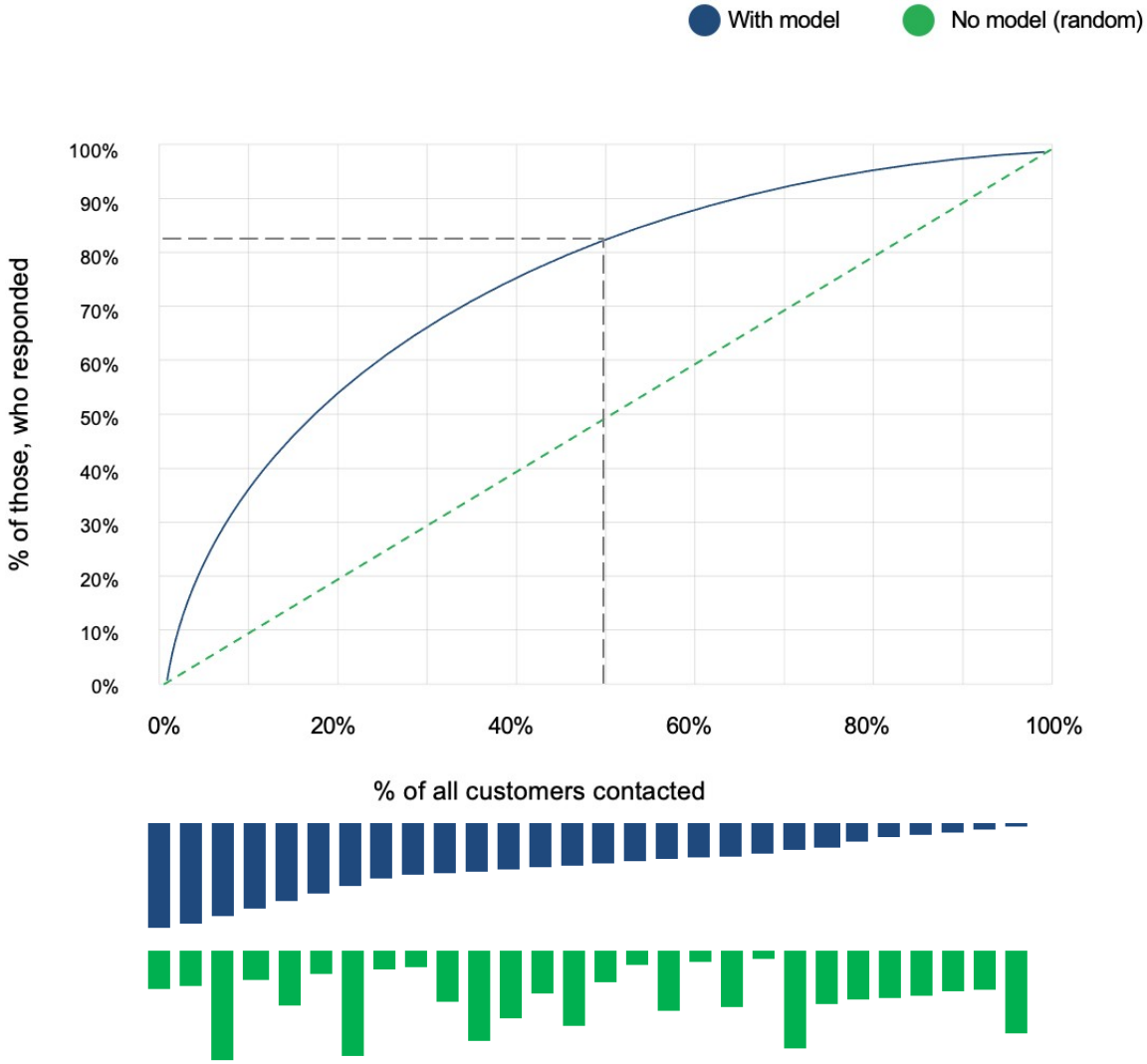


Lift for 50% of cases

50% of cases = 2nd quartile = median

Without model: 50% of positive cases
With model: 80% of positive cases

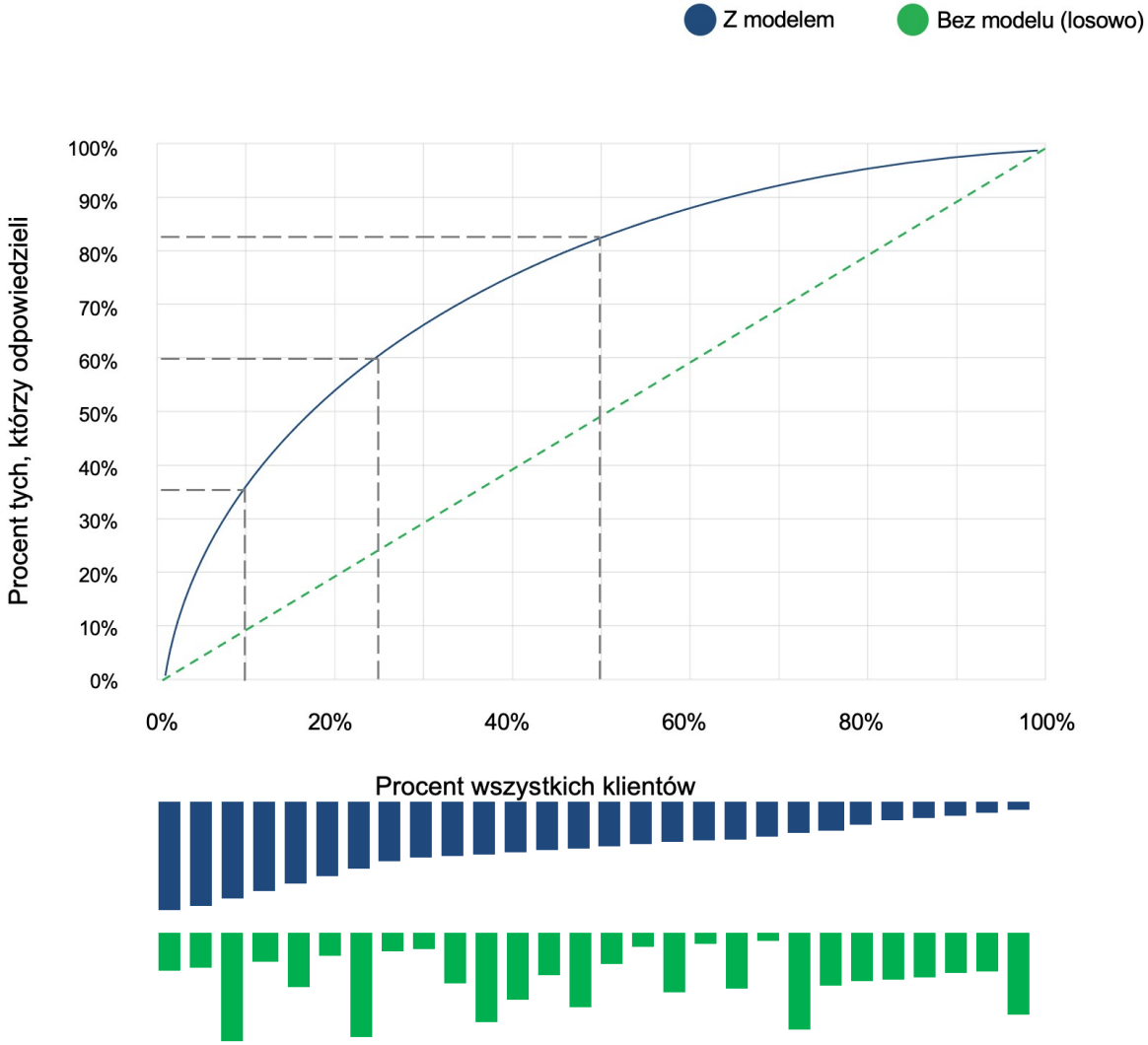
Lift = 0.8/0.5 = 1.6



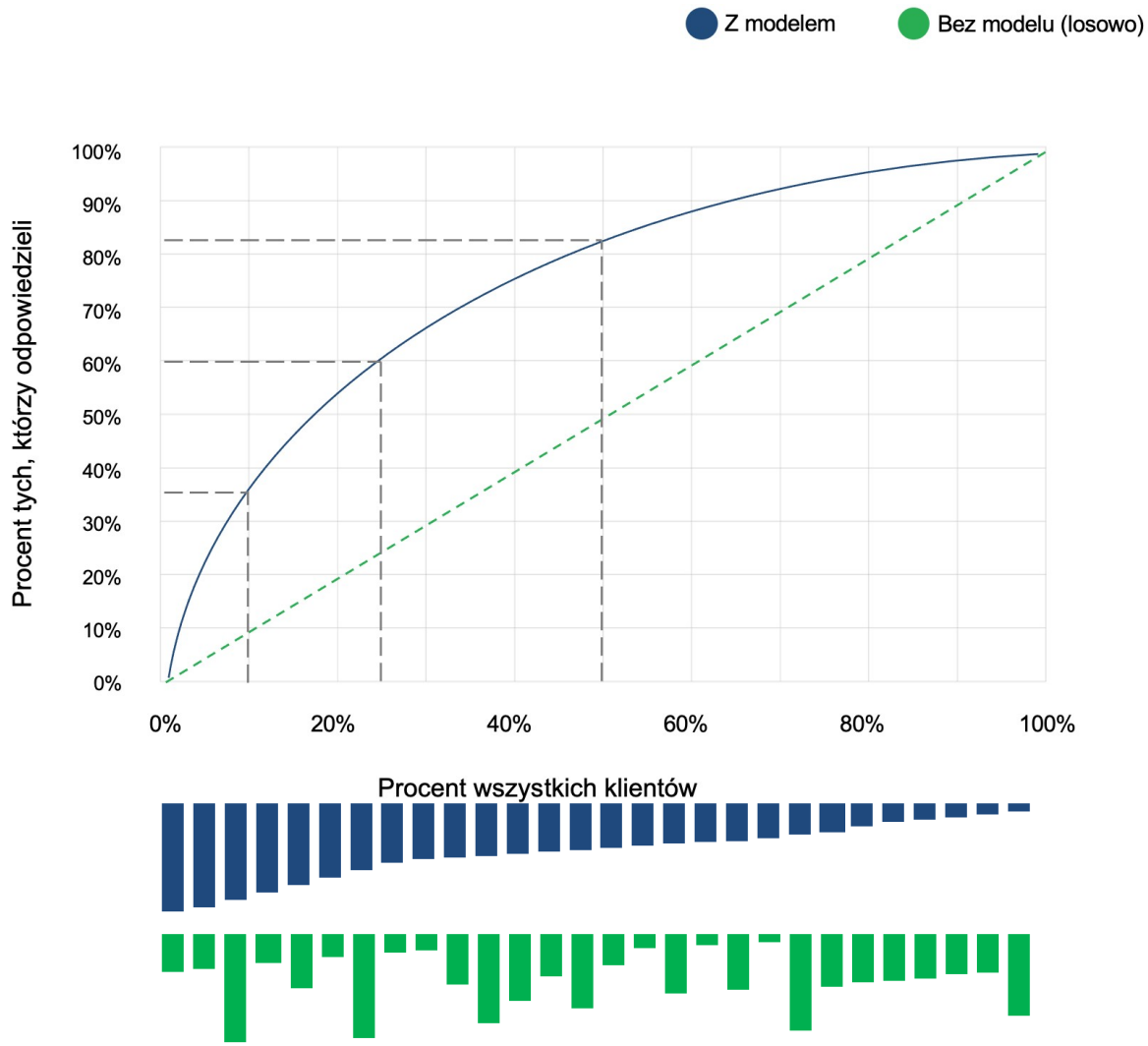
Marketing campaign

Percentage of all cases > Percentage of people we can send letters to

Percentage of positive cases > Percentage of people who responded to the contact

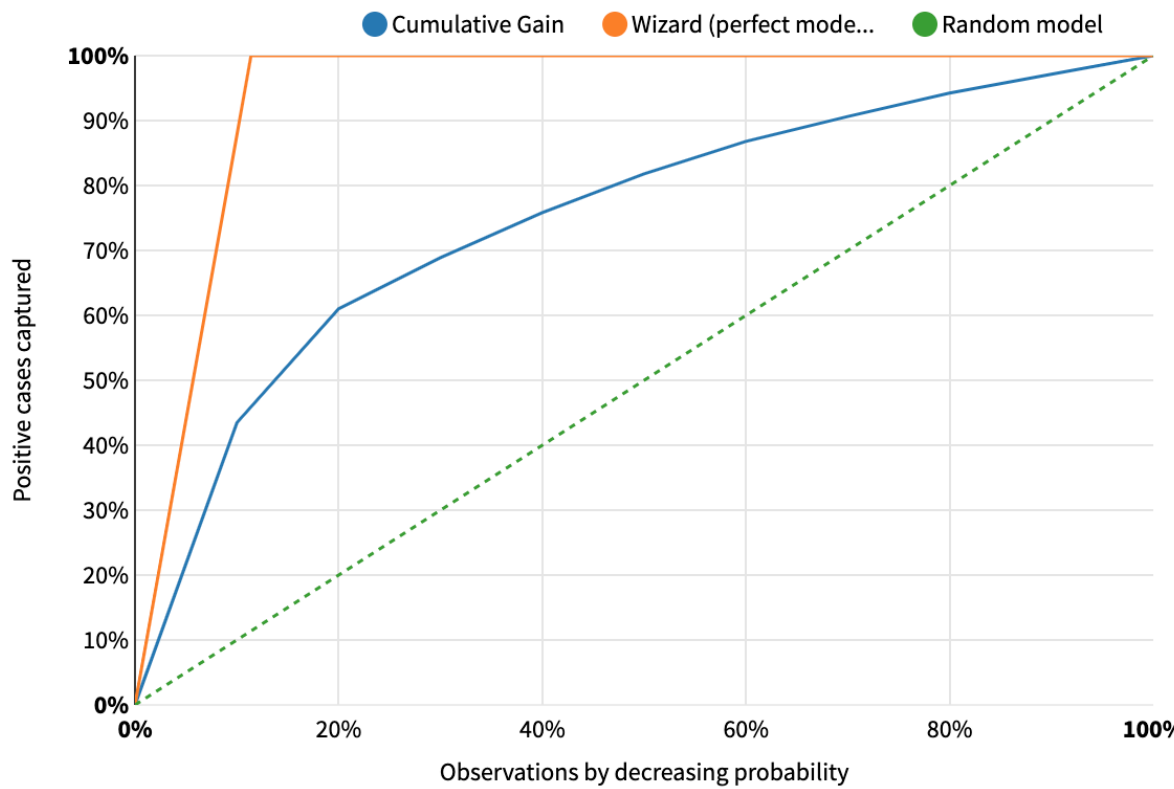


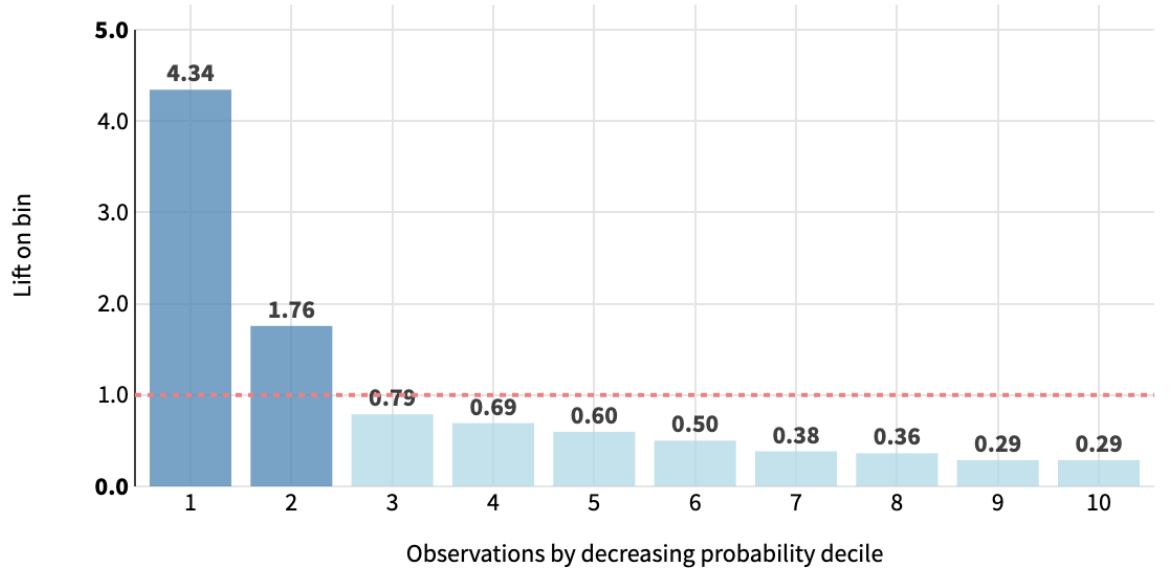
As the number of customers increases, efficiency (lift) decreases



What is the optimal number of customers to contact?

Another form of illustration: deciles





How to do this for CHURN analysis?

Cost matrix

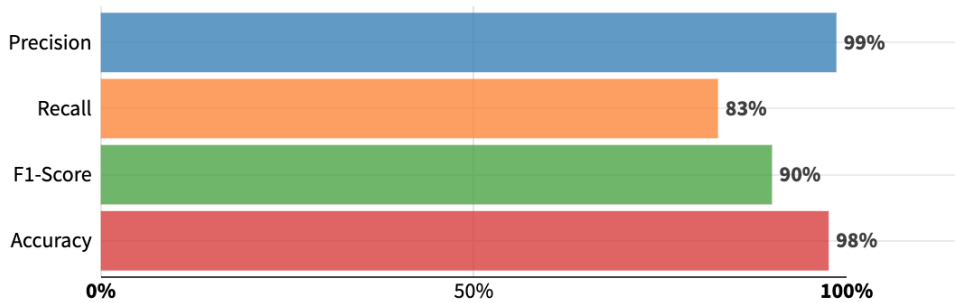
Parameters:

Contact cost: 5 EUR
Cost of promotion: 20 EUR
Profit from acquired customer: 500 EUR

Threshold (cut-off) 0 — 1 | 0.575 | BACK TO OPTIMAL*

Display: Record count

	Predicted 1	Predicted 0	Total
Actually 1	72	15	87
Actually 0	1	578	579
Total	73	593	666



Cost matrix

If model predicts 1	and value is 1	the gain is	475	×	72	=	34,200.00
	but value is 0	the gain is	-25	×	1	=	-25.00
Model predicts 0	and value is 0	the gain is	0	×	578	=	0.00
	but value is 1	the gain is	-500	×	15	=	-7,500.00
	Average gain per record		**40.05**	×	666	=	26,675.00

Example calculation

	Cost	Revenue	Profit
True Positives	15% discount for 33 months (74.41-63.25)*33 = 368.28	Earn 33 months of revenue 63.25*33 = 2087$	1718.72$
True Negatives	If a customer does not churn, it has no effect on our model 0	0	
False Positives	15% discount for 33 months (74.41-63.25)*33 = 368.28		-368.28
False Negatives		Lose 33 months of revenue 63.25*33 = 2087$	-2087

The cost and revenue break down of each classification rate based on the confusion matrix.

The break down of each rate is explained as follows:

- True Positives (Revenue): this is when we give discount to customers we predicted as churn and in response, they are predicted as churn and in response, they no longer churn. We earn 33 months of revenue minus the 15% discount. The total profit is equal 1718.72$
- True Negatives: this is the rate when we predict customers do not churn, so we do not give discounts to them. We earn profits as usual.
- False Positives: this is when we predict they churn and they actually stay. We did give discount to them so we lose profits of the discount. The amount lost is 368.28$.
- False Negatives: this is the rate where we lose 33 months of revenue when we predicted customers stay while they actually churn. 33 months of revenue would be lost since we did not give discount to them, while they can actually change their decision to stay. The total cost of this rate is 2087$.

https://tamhn.medium.com/maximizing-business-impact-with-churn-prediction-using-machine-learning a-practical-approach-d29de2bb4365